

# PHYS-467 Assignment 3

Pietro Giugiaro

December 2022

## Question 2

The atomization energies of the molecules contained in the training set can be displayed in an histogram as shown in Fig.1. The labels of the training set, validation set and test set should be normalized in order to speed up the convergence of the optimization dynamics and to avoid numerical instabilities. In Fig.2 are shown the normalized training set labels.

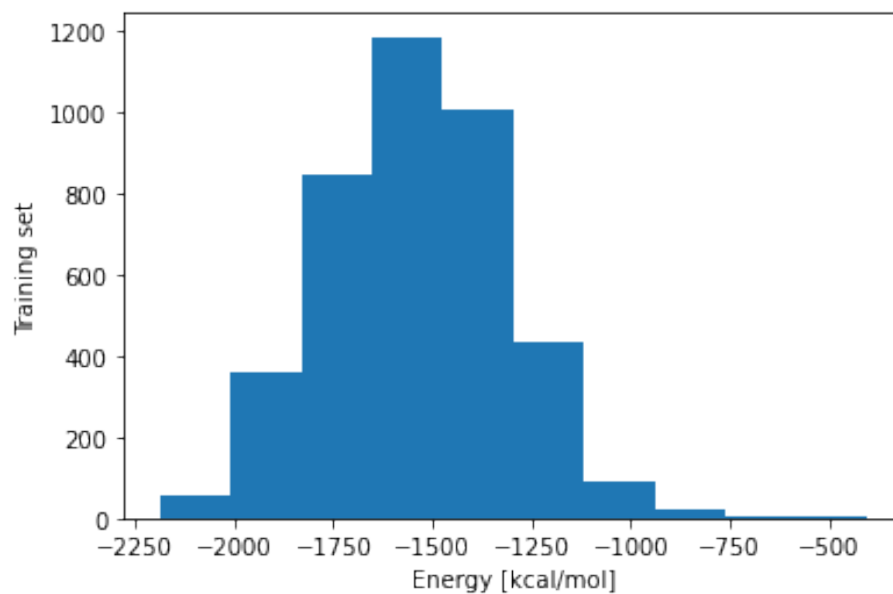


Figure 1: Atomization energies training set

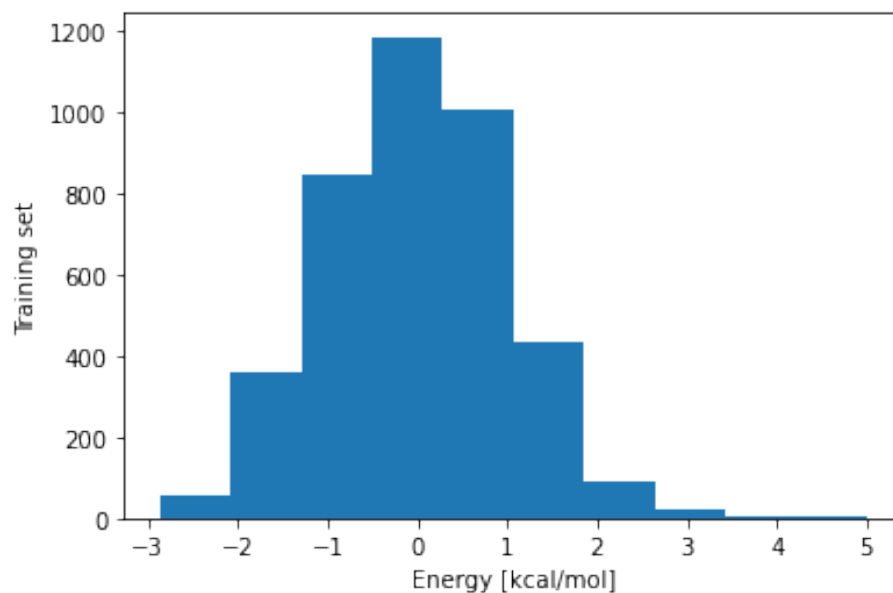


Figure 2: Atomization energies training set normalized

## Question 5

The neural network was trained with four different learning rates, more precisely  $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$ . The optimal value for the learning rate was chosen by evaluating the performance of the neural network on the validation set. More precisely, both the train and the validation losses were plotted over the epochs. As shown in Fig.3, the optimal learning rate is 0.0001, since the validation loss is the the lowest along the epochs.

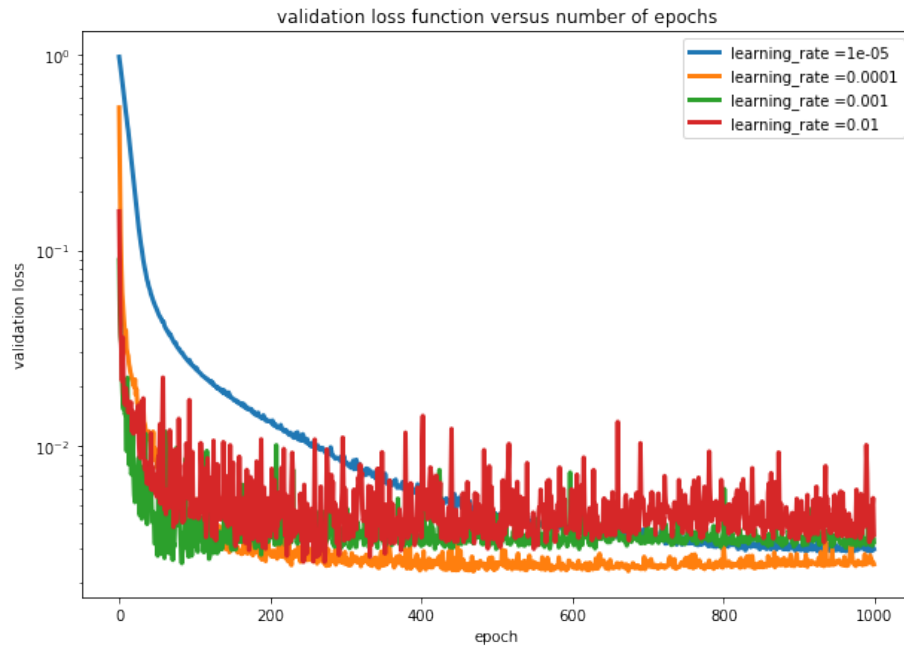


Figure 3: Validation loss function over the training epochs

In Fig.4 is shown the behaviour of the training loss, while in Fig.5 are compared the training and validation losses for the optimal learning rate.

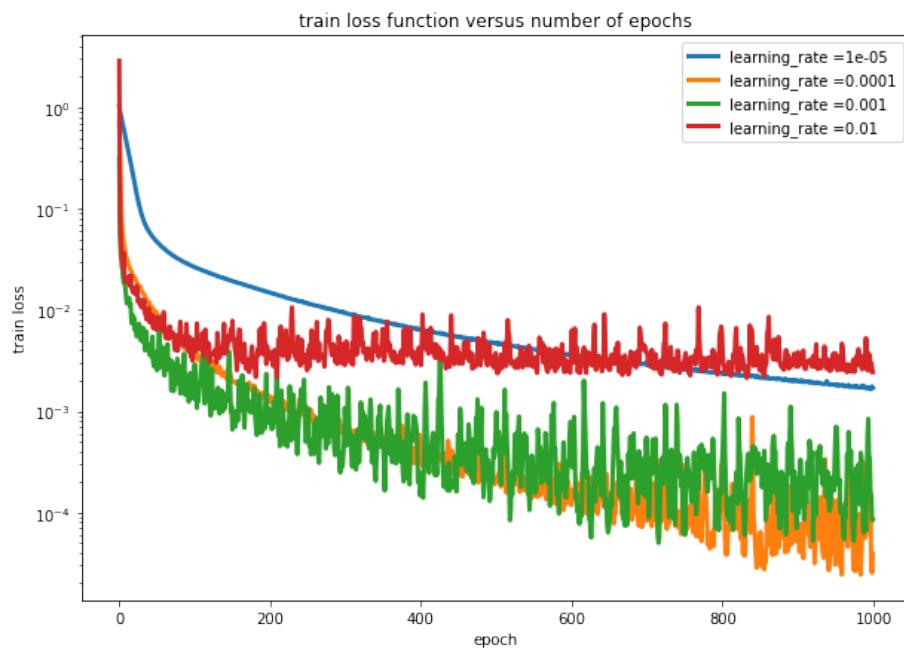


Figure 4: Training loss function over the training epochs

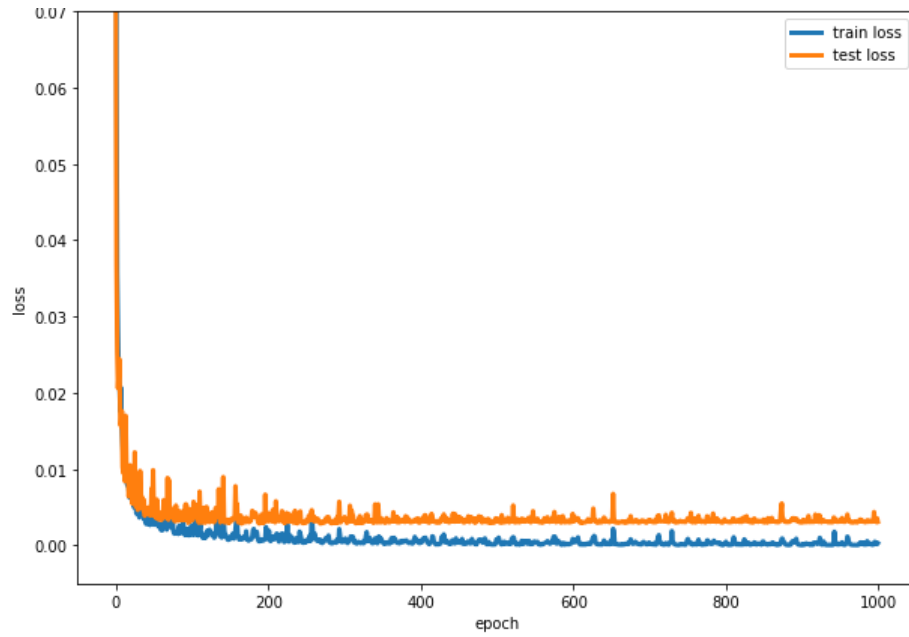


Figure 5: Training and Validation loss function for  $lr = 0.0001$

## Question 6

In Fig.6 is shown the matching between the actual labels and the predicted one in the test set. The root mean square error calculated has approximately value  $RMSE =$

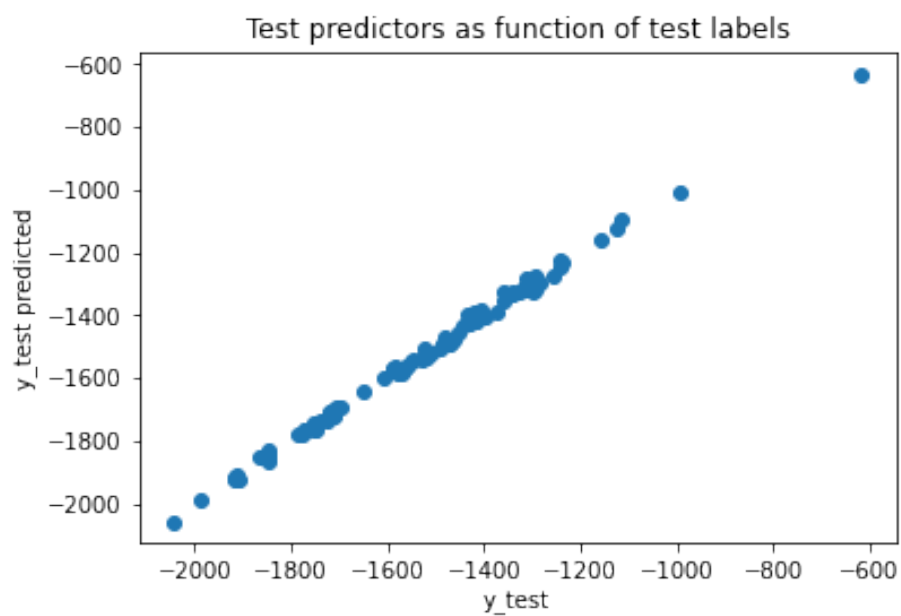


Figure 6: Test set predicted energies *vs* real energies in [*kcal/mol*]

In Fig.6 are compared the predicted labels and the actual labels in the test set.

## Question 7

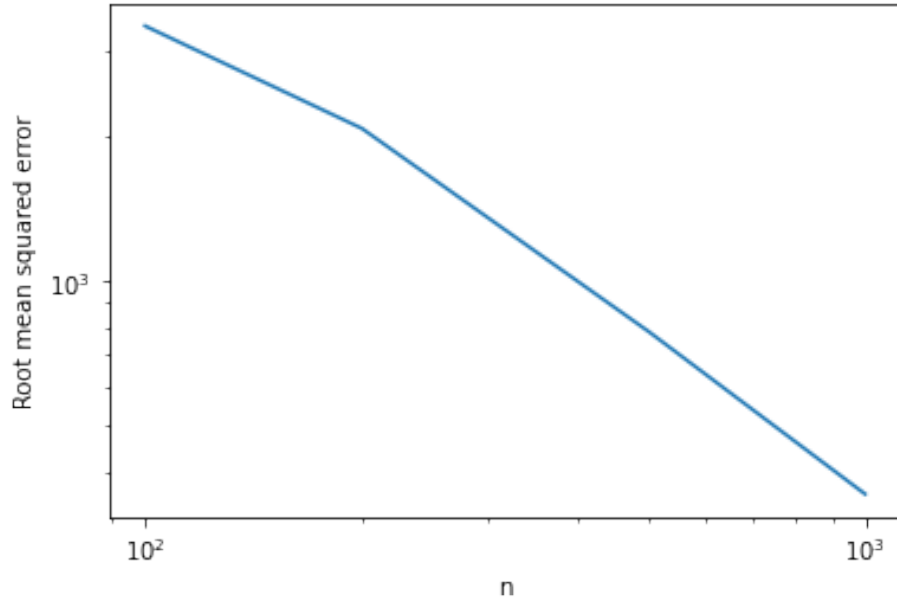


Figure 7:

The neural network was trained again varying the size of the training set  $n \in \{100, 200, 500, 1000\}$ . For each  $n$  the test error was computed and plotted as shown in Fig.7. As expected we see that the MSE decreases as the the number of training samples increases. In a logarithmic scale the behaviour of the test error as a function of  $n$  seem to be linear. Therefore, the best fitting curve for the test error would be of the form  $\epsilon_T = An^{-\beta}$ , which corresponds to  $\log(\epsilon_T) = \log(A) - \beta \log(n)$ . In Fig.8 is shown the fitting of the test error with a function of this kind, with optimal parameters  $A =$  and  $\beta =$ .

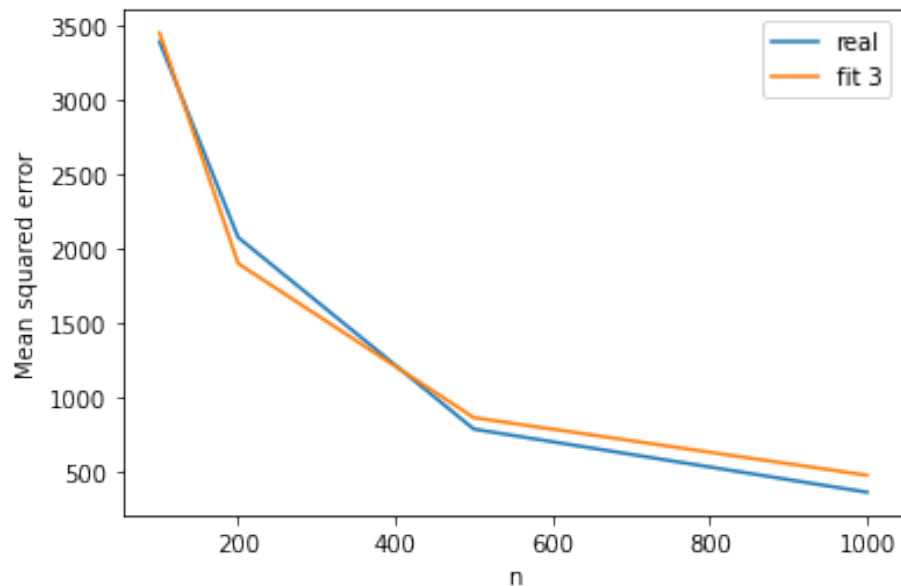


Figure 8:

By using this function to describe the behaviour of the test error is possible to estimate the value of  $\epsilon_T$  for  $n = 4000$ . More precisely, the RMSE for  $n = 4000$  is VALUE, which is comparable with the value obtained in Question 6, obtained with the whole training set. The other fitting curves are actually shown in FIG. The fitting for the first function was not possible by using "curvefit".

## Consulted

Alessandro D'Urso